

REVIEW ARTICLE

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Pheochromocytoma and Paraganglioma

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This article was updated on August 8, 2019, at NEJM.org.

N Engl J Med 2019;381:552-65.

DOI: 10.1056/NEJMra1806651

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PHEOCHROMOCYTOMA FASCINATES AND, AT TIMES, CONFOUNDS CLINICIANS. The symptoms due to hypersecretion of catecholamines can mimic more than 30 medical disorders.¹ This rare tumor can be lethal if left undiagnosed. Thus, rapid recognition is vital. Yet selecting a good approach to biochemical testing and selecting imaging studies for purposes of localization can be confusing and complex. Although surgical resection is the treatment of choice, the preparation for and timing of surgery are uncertain; furthermore, selection of the appropriate operative technique for a given tumor may be controversial. Finally, recent progress in understanding the genetic basis for these tumors and how to manage a specific case according to the gene involved adds to the complexity. Here, we highlight recent advances in the diagnosis and management of this rare neoplasm.

HISTORY, TERMINOLOGY, AND HISTOLOGIC FEATURES

Max Schottelius (1849–1919), a pathologist in Freiburg, Germany, was the first person to describe the histopathological features of pheochromocytoma; in 2017, Schottelius was finally profiled.² He provided the histologic description of bilateral adrenal tumors in an 18-year-old woman who died in 1886 after a long history of panic attacks, tachycardia, and sweating.³ This report was translated into English and published in the series “Classics in Oncology” in 1984.⁴ In 2007, we reported that this patient was the first described to have multiple endocrine neoplasia type 2 (MEN-2); descendants of her brother were found to carry a rearranged during transfection (RET) mutation.⁵ The adrenal tumors in this patient were described by Schottelius as chromaffin tumors on the basis of staining with Müller’s solution. The term pheochromocytoma dates back to 1912, when it was used by the German pathologist Ludwig Pick.⁶

In the 2017 World Health Organization (WHO) classification, pheochromocytoma is an adrenal tumor, and paraganglioma is an extraadrenal tumor; since the two tumor types cannot be differentiated on the basis of histologic findings (Fig. 1), anatomical location is used to distinguish between them.^{7,8} Tumor growth shows the so-called zellballen pattern, consisting of well-developed tumor cells showing nested growth, with an intervening stromal component of fibrovascular tissue and peripheral sustentacular cells. Immunohistochemical examination shows chief cells stained with chromogranin and sustentacular cells stained with S100.⁸

CLINICAL PRESENTATION AND DIAGNOSIS

The incidence of pheochromocytoma and paraganglioma is about 0.6 cases per 100,000 person-years.⁹ A broad spectrum of potential presenting symptoms includes the classic triad of headaches, palpitations, and profuse sweating.¹⁰ Symp-

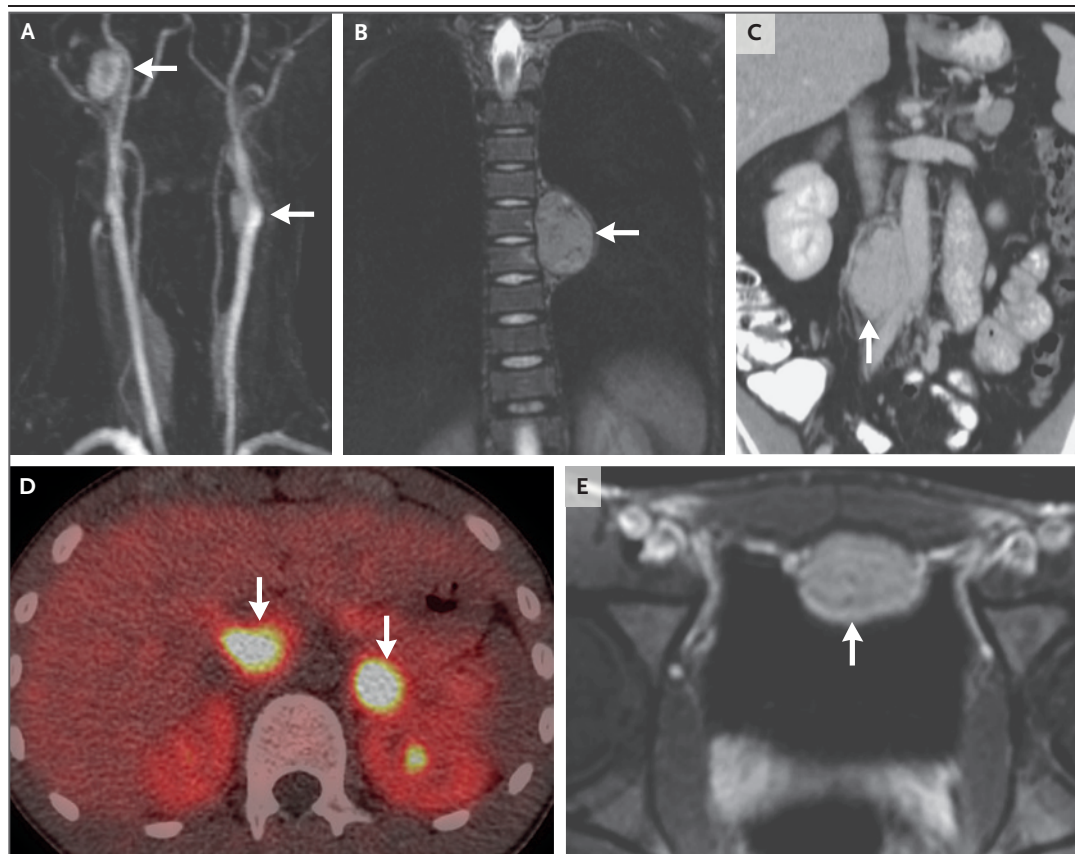


Figure 1. Anatomical Location of Pheochromocytoma and Paraganglioma.

Panel A shows a tumor in the right carotid body (arrows), seen in the frontal view on magnetic resonance imaging (MRI) with gadolinium contrast medium. Panel B shows a thoracic paravertebral paraganglioma on the left side (arrow), seen in the frontal view on MRI with gadolinium contrast medium. Panel C shows a retroperitoneal paraganglioma (arrow) in the frontal view on MRI with gadolinium contrast medium. Panel D shows a left adrenal pheochromocytoma and a right retroperitoneal paraganglioma (arrows) in the axial plane on ^{18}F -labeled L-dihydroxy-phenylalanine–positron-emission tomography–computed tomography (^{18}F -L-DOPA–PET–CT). Panel E shows a paravesicular pelvic paraganglioma on the anterior wall (arrow) in the axial plane on MRI with gadolinium contrast medium.

toms of anxiety and panic attacks are common in the general population, and identifying the rare patient with pheochromocytoma or paraganglioma is a challenge for clinicians. However, with the widespread use of cross-sectional imaging, the incidental discovery of an adrenal mass is becoming common.¹¹ In addition, asymptomatic cases of pheochromocytoma and paraganglioma are increasingly being discovered on the basis of family and germline-mutation testing.

The approach to making a diagnosis may differ from center to center, but a consensus approach is shown in Table 1. The diagnosis of pheochromocytoma or paraganglioma requires both proof of excessive release of catecholamines

and anatomical documentation of the tumor. An increase in plasma fractionated metanephrines (metanephrine and normetanephrine) has a mean sensitivity of 97% and a specificity of 93% across 15 studies.¹² In contrast, measurement of fractionated catecholamines (epinephrine, norepinephrine, and dopamine) is less sensitive, but clearly elevated values (>2 times the upper limit of the normal range) are also diagnostic.^{13,14} However, mild elevations in the levels of fractionated metanephrines and catecholamines in plasma and urine are common in persons who do not have pheochromocytoma. Medications (e.g., tricyclic antidepressants, antipsychotic agents, serotonin-reuptake or norepinephrine-

Table 1. Approach to Pheochromocytoma and Associated Syndromes According to the Clinical Scenario.*

Clinical Scenario	Initial Biochemical Testing and Imaging	Follow-up Biochemical Testing and Imaging
Signs and symptoms on presentation (e.g., resistant hypertension or paroxysms of hypertension, palpitations, perspiration, headaches, and markedly elevated metanephrines or catecholamines)	Perform abdominal contrast-enhanced CT or MRI; if abdominal imaging is negative, consider MRI of skull base, neck, chest, and pelvis	Measure metanephrines postoperatively and then annually; if bilateral pheochromocytomas were removed with cortical-sparing surgery, document normal glucocorticoid secretory function with cosyntropin-stimulation test
Incidentally discovered adrenal or retroperitoneal mass with attenuation >10 Hounsfield units on unenhanced CT	If levels of metanephrines are clearly elevated, perform contrast-enhanced CT or MRI; if mass is >10 cm in diameter or is extraadrenal, search for additional paragangliomas or metastatic disease with ¹²³ I-MIBG scintigraphy, ⁶⁸ Ga-DOTATATE-PET-CT, or ¹⁸ F-FDG-PET-CT	If a pheochromocytoma or paraganglioma was resected, measure metanephrines postoperatively and then annually
Patient identified as carrier of disease-causing mutation		
<i>RET</i> mutation	Measure metanephrines and perform abdominal MRI; measure serum calcitonin and calcium; seek endocrine surgery consultation if thyroid gland not previously resected	Measure serum calcitonin, metanephrines, and serum calcium annually
<i>VHL</i> mutation	Measure metanephrines; perform MRI of the brain, spinal cord, and abdomen; perform ophthalmoscopy	Measure metanephrines yearly; perform MRI of the brain, spinal cord, and abdomen; perform ophthalmoscopy; if no tumor found, monitor every 2 or 3 years
<i>SDHA</i> , <i>SDHB</i> , or <i>SDHD</i> mutation	Perform MRI of skull base and neck, thorax, retroperitoneum, and pelvis; alternatively, perform ⁶⁸ Ga-DOTATATE-PET-CT; also measure metanephrines	Measure metanephrines yearly; if a pheochromocytoma or paraganglioma was removed, perform MRI of the surgical region annually for yr 1–3; for body areas that had no tumors, perform MRI every 3 yr
<i>SDHC</i> or <i>SDHAF2</i> mutation	Measure metanephrines; perform MRI of skull base and neck or ⁶⁸ Ga-DOTATATE-PET-CT	If a paraganglioma or pheochromocytoma was removed, perform MRI of the surgical region annually for yr 1–3; for body areas that had no tumors, perform MRI every 3 to 5 yr
<i>MAX</i> or <i>TMEM127</i> mutation	Measure metanephrines; perform MRI of the abdomen or ⁶⁸ Ga-DOTATATE-PET-CT	Measure metanephrines yearly; if a pheochromocytoma or paraganglioma was removed, perform MRI of the surgical region annually for yr 1–3; perform MRI of the abdomen every 3 yr
Neurofibromatosis type 1	Measure metanephrines	If hypertension or clinical symptoms develop, measure metanephrines

* The term metanephrines refers to metanephrine and normetanephrine. CT denotes computed tomography, ¹⁸F-FDG ¹⁸F-fluorodeoxyglucose, ⁶⁸Ga-DOTATATE ⁶⁸Ga-labeled 1,4,7,10-tetraazacyclododecane-1,4,7,10-tetraacetic acid-octreotate, ¹²³I-MIBG ¹²³I-labeled metaiodobenzylguanidine, MRI magnetic resonance imaging, and PET positron-emission tomography.

reuptake inhibitors, and levodopa) could cause elevations in endogenous catecholamines, and clinical circumstances (e.g., acute illness) could lead to false positive test results. To effectively screen for catecholamine-secreting tumors, tricyclic antidepressants and other psychoactive agents should be tapered and discontinued at least 2 weeks before any hormonal assessments are performed.

For medical imaging of pheochromocytoma or paraganglioma (Table 1), three scenarios must

be considered: first, typical symptoms combined with clearly elevated metanephrine or catecholamine levels; second, incidental detection of an adrenal or retroperitoneal mass; and third, a germline mutation in a susceptibility gene on molecular genetic testing and the associated syndrome. In scenario 1, the tumor has to be localized by means of either contrast-enhanced computed tomography (CT) or T2-weighted magnetic resonance imaging (MRI). Standard imaging includes the entire retroperitoneum,

since nearly all extraadrenal, catecholamine-secreting tumors are located in the retroperitoneum rather than in the pelvis or thorax. In scenario 2, CT without contrast material is important, because when CT attenuation is 10 Hounsfield units or less, a lipid-rich mass is present, which rules out the diagnosis of pheochromocytoma or paraganglioma, and biochemical testing is not needed.¹⁵ For masses with CT attenuation of more than 10 Hounsfield units, biochemical testing must be performed. When biochemical testing is abnormal, cross-sectional imaging with contrast-enhanced CT or MRI is indicated. For scenario 3, see the discussion below on the care of asymptomatic germline mutation carriers and relatives.

Once a pheochromocytoma or paraganglioma has been identified, the need for additional total-body imaging is questionable.¹⁶ Functional imaging (e.g., scintigraphy with ¹²³I-labeled meta-iodobenzylguanidine [MIBG] or positron-emission tomography [PET]–CT with ⁶⁸Ga-labeled 1,4,7,10-tetraazacyclododecane–1,4,7,10-tetraacetic acid–octreotate [DOTATATE] or ¹⁸F-labeled L-dihydroxyphenylalanine [L-DOPA]) is very effective in localizing pheochromocytomas and paragangliomas (Fig. 1).¹⁷ Furthermore, the results of ⁶⁸Ga-DOTATATE–PET–CT correlate with biochemical measurements.¹⁸ The main indication for functional imaging is the search for metastatic disease or identification of multiple chromaffin tumors.

In contrast, head-and-neck paragangliomas are usually manifested as painless, slowly growing masses, mainly as carotid-body tumors and vagal paragangliomas, or with conductive hearing loss and pulsatile tinnitus caused by jugulotympanic paragangliomas. Lower-cranial-nerve deficits are frequently seen in patients with advanced head-and-neck paragangliomas. Catecholamine hypersecretion is rarely found in such patients.^{19,20}

TREATMENT

Surgical resection of pheochromocytoma or paraganglioma is the cornerstone of therapy. Most of these tumors are resected on the basis of biochemical and CT or MRI documentation. The major questions concern the timing of surgery and the surgical approach.

Combined α - and β -adrenergic blockade is

standard treatment for patients with pheochromocytoma in order to control blood pressure and prevent intraoperative hypertensive crises.²¹ Adrenergic blockade is typically accomplished with either a nonselective or a selective α -adrenergic receptor antagonist (e.g., nonselective α -adrenergic blockade with phenoxybenzamine, started at a dose of 10 mg by mouth two times a day and increased to up to 30 mg three times a day with adjustment for low-normal blood pressure for age; or selective α_1 -adrenergic blockade with doxazosin, started at a dose of 1 mg by mouth daily and increased to up to 10 mg twice daily as needed to achieve target blood-pressure levels), usually started at least 7 days before surgery.^{1,22} Adrenergic blockade should be accompanied by a high-sodium diet (e.g., 5000 mg per day) and generous fluid intake (e.g., 2.5 liters per day).¹

The β -adrenergic antagonist (e.g., oral administration of extended-release metoprolol at a starting dose of 25 mg by mouth once daily and increased to up to 100 mg twice daily as needed for a target average heart rate of 80 beats per minute) should be administered to control tachycardia after α -adrenergic blockade has been effective in normalizing blood pressure, because with β -adrenergic blockade alone, severe hypertension or cardiopulmonary decompensation may occur as a result of unopposed α -adrenergic stimulation.²² However, postoperative sustained hypotension can be a complication of preoperative adrenergic blockade. In 2017, the concept of adrenergic blockade-free management was explored. A prospective study involving 110 patients who received preoperative treatment with α -adrenergic blockade and 166 patients who did not revealed no relevant differences regarding maximal intraoperative systolic pressure, hypertensive episodes, or major complications.²³ In a patient who has not had adverse effects in association with long-term, episodic blood-pressure spikes without α -adrenergic blockade, it seems unlikely that decompensation would occur in the few days before surgery, and nitroprusside-guided blood-pressure control during anesthesia may be justifiable. This opens the avenue for operation without further delay. There is no consensus in support of this approach, not even among us, and the 2014 pheochromocytoma guidelines issued by the Endocrine Society, as well as many experts, continue to recommend preoperative adrenergic blockade for all patients.²²

Until 1996, the surgical approach for pheochromocytoma was open laparotomy and removal of the entire adrenal gland with the tumor.²⁴ On the basis of anecdotal evidence from many centers in the United States, Europe, and Asia, this procedure is still frequently performed. In 1996, Gagner and colleagues removed a pheochromocytoma endoscopically.²⁵ Over the subsequent decades, endoscopic technology with transabdominal or retroperitoneal access replaced the open surgical approach and became standard practice because of the shorter operative time, fewer intraoperative and postoperative complications, and shorter hospital stay.²⁶ Although it has been shown that pheochromocytomas with a diameter of 5 cm or more can be safely removed endoscopically,²⁷ their safe removal is still a matter of debate and should be considered on an individual basis according to tumor characteristics and surgical expertise. For bilateral adrenal pheochromocytoma, the concept of organ-sparing surgery was introduced in 1999 to avoid a postoperative requirement for glucocorticoid and mineralocorticoid replacement.²⁸ Approximately one third of one adrenal gland is sufficient for normal glucocorticoid and mineralocorticoid secretion.²⁹ Much like retroperitoneal tumors, tumors in unusual locations such as pelvic paravascular and thoracic paravertebral regions have been removed with the use of minimally invasive surgery³⁰ (see the video, available with the full text of this article at NEJM.org; and the illustrative case in the Supplementary Appendix, available at NEJM.org).

For a patient with a head-and-neck paraganglioma, finding the best treatment option often represents a challenge, and an individualized interdisciplinary approach is essential. Treatment options include surgery, stereotactic radiosurgery, external radiation therapy, and a wait-and-scan strategy. Determining the exact location and presence of tumor extension, as well as staging before any kind of treatment, is essential. Carotid-body tumors are most frequently classified with the use of the Shamblin classification, whereas the Fisch classification is used for jugulotympanic paragangliomas, both of which guide surgical management.^{31,32} Surgical resection is the only potentially curative treatment option. With advanced cervical paragangliomas (Shamblin class III) and jugular paragangliomas (Fisch class C and D), lower-cranial-nerve deficits are frequent-

ly seen after surgery. Patients with these advanced tumors may have less treatment-related morbidity with the nonsurgical treatment options.

SUSCEPTIBILITY GENES

The *RET* proto-oncogene was identified in 1993 as a risk factor for pheochromocytoma. Since then, at least 18 additional susceptibility genes have been reported.³³ In parallel with these investigations, gene-specific clinical data have been studied to guide clinical management strategies. Here, we delineate the most extensively studied clinical phenotypes (e.g., tumor location, number of tumors, and patient age at tumor manifestation) of the 10 currently clinically relevant syndromes (Table 2) that have been identified: MEN-2, caused by germline mutations of the *RET* proto-oncogene³⁴; von Hippel–Lindau disease, caused by mutations in the *VHL* tumor-suppressor gene³⁵; neurofibromatosis type 1, caused by mutations in the *NF1* tumor-suppressor gene³⁶; paraganglioma syndromes 1 through 5, caused by mutations of the succinate dehydrogenase genes *SDHD* (syndrome 1),³⁷ *SDHAF2* (syndrome 2),³⁸ *SDHC* (syndrome 3),³⁹ *SDHB* (syndrome 4),⁴⁰ and *SDHA* (syndrome 5)⁴¹; and the hereditary pheochromocytoma syndromes caused by mutations in the genes encoding transmembrane protein 127 (*TMEM127*)⁴² and MYC-associated factor X (*MAX*).⁴³ Other susceptibility genes, which have not yet undergone rigorous genotype–clinical outcome evaluation, include *EGLN1* (*PHD2*), *EGLN2* (*PHD1*), *KIF1B*, *IDH1*, *HIF2A*, *MDH2*, *FH*, *SLC25A11*, and *DNMT3A*.^{44–53} Various publicly available gene-variation databases that are relevant to pheochromocytoma or paraganglioma susceptibility genes (e.g., LOVD-based *SDHx* variant database [www.LOVD.nl]) may be useful for clinicians and researchers.⁵⁴

PHEOCHROMOCYTOMA-ASSOCIATED SYNDROMES

The clinical features of pheochromocytoma-associated syndromes (Table 2) have been known for more than 100 years, starting with neurofibromatosis 1, von Hippel–Lindau disease, and MEN-2. With the identification of susceptibility genes containing various germline mutations, these and additional syndromes have been defined and differentiated. Unilateral or bilateral pheochro-



A video showing endoscopic removal of a thoracic paraganglioma is available at NEJM.org

Table 2. Characteristics of Pheochromocytoma-Associated Syndromes.*

Gene	Syndrome	Nonchromaffin Tumors	Transmission	Adrenal Tumors	Head and Neck Tumors	Extraadrenal Tumors†	Multiple Tumors	Metastatic Tumors‡	Family History§
<i>frequency (percent)</i>									
VHL	VHL	Retinal and CNS hemangioblastomas, RCC, pancreatic neuroendocrine tumor, ELST	Autosomal dominant	>50	<1	10–24	>50	1–9	25–50
NF1	NF1	Cutaneous neurofibromas, malignant peripheral-nerve-sheath tumor, breast cancer	Autosomal dominant	>50	<1	1–9	25–50	1–9	10–24
RET	MEN-2	Medullary thyroid carcinoma, hyperparathyroidism	Autosomal dominant	>50	<1	<1	>50	<1	25–50
SDHA	PGL5	Rarely also pituitary adenoma, GIST, RCC	Autosomal dominant	25–50	25–50	25–50	1–9	1–9	1–9
SDHB	PGL4	Rarely also pituitary adenoma, GIST, RCC	Autosomal dominant	25–50	25–50	25–50	10–24	25–50	10–24
SDHC	PGL3	Rarely also pituitary adenoma, GIST	Autosomal dominant	1–9	>50	<1	10–24	Not reported	10–24
SDHD	PGL1	Rarely also pituitary adenoma, GIST, RCC	Autosomal dominant, maternal imprinting	10–24	>50	10–24	>50	1–9	25–50
SDHAF2	PGL2		Autosomal dominant, maternal imprinting	1–9	>50	Not reported	>50	Not reported	>50
MAX	No name	Rarely also RCC	Autosomal dominant	>50	<1	1–9	>50	1–9	25–50
TMEM127	No name		Autosomal dominant	>50	1–9	<1	25–50	10–24	1–9

* For multiple endocrine neoplasia type 2 (MEN-2), von Hippel–Lindau disease (VHL), and neurofibromatosis type 1 (NF1), the frequencies of the characteristics shown are for patients with chromaffin tumors, since such tumors do not develop in all patients with these syndromes. CNS denotes central nervous system, ELST endolymphatic-sac tumor of inner ear, GIST gastrointestinal stromal tumor, HPT hyperparathyroidism, PGL paraganglioma syndrome (PGL1 through PGL5 denote paraganglioma syndromes 1 through 5), and RCC renal-cell carcinoma.

† These tumors consist of retroperitoneal, pelvic, and thoracic tumors.

‡ These tumors consist of metastatic pheochromocytoma and paraganglioma.

§ Shown is the frequency of a family history of components of the given syndrome.

mocytomas develop in up to 50% of patients with these syndromes.^{55,56} Once the pathogenic *RET* mutation is identified in a patient with pheochromocytoma, clinicians must be aware that virtually all mutation carriers will have medullary thyroid carcinoma, whereas 20% of patients with MEN-2A, but not MEN-2B, will have hyperparathyroidism. It would be rare for a patient with MEN-2B (characterized chiefly by the *RET* p.M918T mutation) to first present with pheochromocytoma, since the typical components are manifested relatively early, even in infancy, such as early-onset medullary thyroid carcinoma; ganglioneuromas typically involving the tongue, lips, and eyelids; skeletal deformities (e.g., kyphoscoliosis and marfanoid habitus); joint laxity; and intestinal ganglioneuromatosis.⁵⁶

By contrast, in patients with von Hippel–Lindau disease, especially type 2, pheochromocytomas and paragangliomas are common, may be part of the patient's initial presentation, even in early infancy, and are often associated with mutations in codons 98, 161, and 167.^{33,57,58} Missense *VHL* mutations predict von Hippel–Lindau disease type 2. Truncating *VHL* mutations are often associated with renal-cell carcinoma and in rare cases are associated with pheochromocytoma (von Hippel–Lindau disease type 1). Other features of von Hippel–Lindau disease include retinal and central nervous system (CNS) hemangioblastomas and pancreatic neuroendocrine tumors.⁵⁹

Neurofibromatosis 1 is characterized by neurofibromas, café au lait spots, axillary freckling, iris hamartomas (Lisch nodules), bony abnormalities, CNS gliomas, macrocephaly, and cognitive deficits, but pheochromocytomas and paragangliomas are present in only 1 to 3% of affected persons.^{60–62} Patients with mutations of *SDHx*, *TMEM127*, or *MAX* generally have only pheochromocytomas and paragangliomas. A minority of *SDHx* mutation carriers have had gastrointestinal stromal tumors (GISTs), renal-cell carcinomas, or pituitary adenomas.^{63–69} The autosomal dominant Carney–Stratakis syndrome is characterized by the association of pheochromocytoma, paraganglioma, or both with GISTs, and the so-called 3PAs syndrome (pheochromocytoma, paraganglioma, and pituitary adenoma) is associated with *SDHx* mutations.^{70,71}

The concept of genetic predisposition may be confusing to patients and their families. Inheri-

Figure 2 (facing page). Frequencies of Gene Mutations in Patients with Pheochromocytoma or Paraganglioma.

Data are from the joint Freiburg, Germany, and Padua, Italy, registries, which contain information about more than 3300 patients with chromaffin tumors.^{7,55,61,73,74}

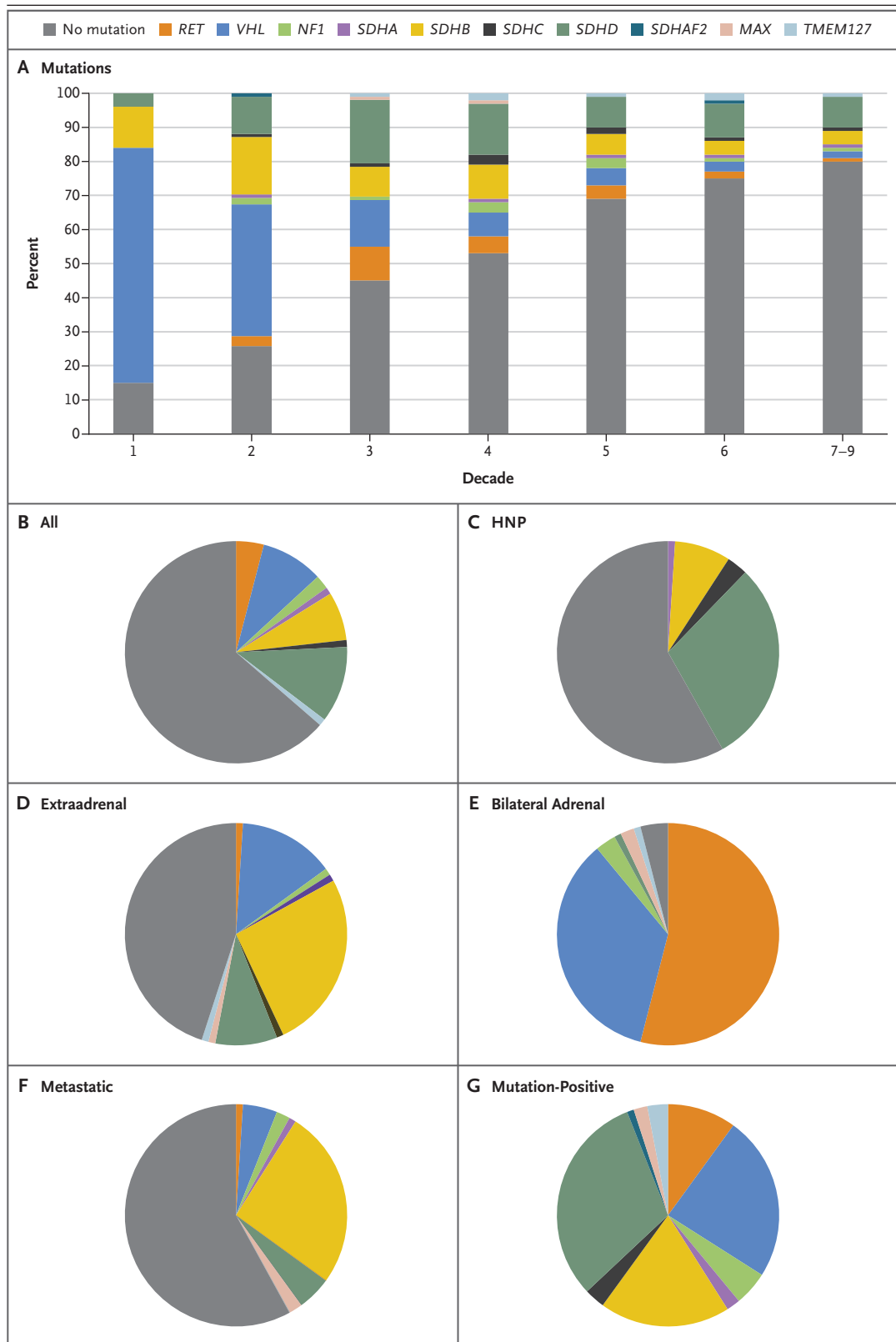
Panel A shows the relative percentages of germline mutations among all mutation carriers, according to decade of age at presentation. In Panels B through G, the relative percentages of patients with mutations in various genes and patients without mutations are shown in pie charts. The overall percentages of mutation carriers are shown in Panel B; the percentages for head-and-neck paragangliomas (HNP), extraadrenal tumors, bilateral adrenal tumors, and metastatic tumors are shown in Panels C through F, respectively. In Panel G, the pie chart for mutation-positive tumors shows the relative frequencies of mutations in susceptibility genes among mutation carriers. Proportions of less than 1% are not shown.

tance is autosomal dominant for pheochromocytoma-predisposing germline mutations. This means that offspring of a carrier have a 50% likelihood of inheriting the parent's mutation. However, *SDHD* and *SDHAF2* are maternally imprinted (the gene is not expressed from the maternal allele). This means that inheriting a mutation from the mother only rarely results in tumor development,⁷² which explains why one or more generations appear to be skipped.

CLINICAL HALLMARKS OF HEREDITARY PHEOCHROMOCYTOMA AND PARAGANGLIOMA

In addition to family history, classic hallmarks of hereditary pheochromocytoma and paraganglioma include an early age at onset, extraadrenal and multiple primary tumors, and associated non-paraganglial tumors (Table 2 and Fig. 2). The age at diagnosis is about 15 years younger for syndromic pheochromocytomas and paragangliomas than for sporadic cases, with von Hippel–Lindau disease diagnosed at the earliest age.⁷⁴ Paraganglioma occurring in an unusual location, such as the organ of Zuckerkandl, thorax, or urinary bladder, should suggest the possibility that the patient has a syndrome associated with the tumor.

Anatomical locations of pheochromocytomas and paragangliomas differ widely among the syndromes (Table 2). Adrenal pheochromocytomas occur almost exclusively in patients with



germline mutations in *RET*. In contrast, in patients with mutations in *VHL*, *NF1*, *MAX*, and *TMEM127*, pheochromocytomas are common, but retroperitoneal paragangliomas are also observed. Patients with *SDHx* mutations frequently have head-and-neck paragangliomas, but pheochromocytomas and retroperitoneal paragangliomas are seen mainly in carriers of *SDHD*, *SDHB*, and *SDHA* mutations. The rare thoracic paragangliomas are associated mostly with mutations in *SDHB*, *SDHD*, or *VHL*. Multiple primary tumors often occur in patients with germline mutations in *RET*, *VHL*, *SDHD*, or *MAX*.

METASTATIC PHEOCHROMOCYTOMA

The diagnosis of malignant pheochromocytoma is problematic. Pathologists have correlated histologic features such as growth patterns, mitoses, and atypia of cells and nuclei with malignant biologic features in pheochromocytomas and paragangliomas, and these findings form the basis for scoring systems (e.g., Pheochromocytoma of the Adrenal Gland Scaled Score [PASS] and Grading System for Adrenal Pheochromocytoma and Paraganglioma [GAPP]).^{75,76} However, it remains difficult to predict the clinical behavior of individual tumors, and no single risk-stratification scheme is endorsed or in widespread use.^{77,78} Only metastases are the proof for a malignant pheochromocytoma or paraganglioma, and the updated WHO classification of endocrine tumors has replaced the term “malignant pheochromocytoma” with “metastatic pheochromocytoma.”⁷⁸ Metastases are located where chromaffin tissue is normally not found (e.g., lymph nodes, lung, liver, and bones), and frequently, metastases are not confirmed histologically but are instead documented on nuclear imaging.⁷⁹ When pheochromocytoma is the primary tumor, the typical metastatic sites are bones and lymph nodes, whereas when paraganglioma is the primary tumor, hepatic metastases are more common.⁸⁰ As a cautionary note, recurrent and metastatic disease may occur after intraoperative rupture of the tumor capsule, a complication that can lead to incurable tumor spread.^{81,82}

Treatment options for metastatic pheochromocytoma include surgical resection, use of targeted radiolabeled carriers (e.g., ¹³¹I-MIBG or ⁹⁰Y-DOTATATE and ¹⁷⁷Lu-DOTATATE), thermal

ablation, chemotherapy, and external irradiation. Surgical resection of as much tumor tissue as possible should be the initial approach.⁸³ Most patients with metastatic pheochromocytoma or paraganglioma have sporadic tumors, whereas among patients with heritable pheochromocytoma in whom metastatic disease develops, tumors caused by *SDHB* mutations account for up to 43% of cases, followed by *VHL*, *SDHD*, and *NF1* mutations.⁸⁴ Overall survival and disease-specific survival are surprisingly favorable, at 25 years and 34 years, respectively.⁸⁰

MAJOR NEEDS OF AFFECTED PATIENTS

The key concerns from the patient's perspective include early diagnosis, the effect of germline mutation on treatment, best surgical option, postoperative care guided by genetic features, and appropriate care and surveillance of relatives. Early diagnosis in patients with sporadic pheochromocytomas or paragangliomas relies on keen clinicians who recognize signs and symptoms of catecholamine-secreting tumors. Once the diagnosis is established, it is important to consider when genetic testing in the context of genetic counseling should be performed, since all patients with pheochromocytoma or paraganglioma should undergo genetic analysis. After detection of the mutation, the specific gene guides the tailoring of imaging studies and subsequent medical management (Table 1). For example, imaging studies are performed for medullary thyroid carcinoma if *RET* is mutated; for hemangioblastomas of the eyes and CNS and for tumors of the ears, kidneys, and pancreas if *VHL* is mutated; and for other chromaffin tumors or the rare kidney cancers, pituitary adenomas, or GISTs if *SDHx*, *MAX*, or *TMEM127* is mutated. An important question is whether screening for mutations should be performed before a pheochromocytoma is resected. Bilateral adrenal tumors can occur metachronously, with intervals of more than a decade, especially in patients with *RET* germline mutations.⁵⁵ Thus, operative planning for preservation of sufficient adrenal cortex will be guided by the results of gene testing.

Favorable surgical outcomes include complete tumor resection, no permanent complications,

limited postoperative pain, good cosmetic results, and a short hospital stay. Minimally invasive surgery performed by expert surgeons is key in achieving these goals.

An organized, long-term postoperative care program is essential for patients with germline mutations and should be in the hands of endocrinologists. The goals of such programs (Table 1) include surveillance and early detection of pheochromocytomas and paragangliomas and surveillance for and management of associated nonchromaffin tumors such as medullary thyroid carcinoma and primary hyperparathyroidism in patients with MEN-2; hemangioblastomas of the retina, cerebellum, and spine, along with renal-cell carcinomas and pancreatic neuroendocrine tumors, in patients with von Hippel–Lindau disease; peripheral-nerve-sheath tumors and breast cancer in patients with neurofibromatosis 1; and GISTs, renal-cell carcinomas, and pituitary tumors in patients with *SDHx* and *MAX* mutations.

TESTING OF RELATIVES
AND ASYMPTOMATIC GERMLINE-
MUTATION CARRIERS

All first-degree relatives of a mutation carrier should be offered predictive (or cascade) testing in the context of genetic counseling. Testing of relatives results in a 50% likelihood of identifying mutation carriers, who are typically asymptomatic. These asymptomatic mutation carriers should be offered clinical surveillance and management specific to the gene (Table 1).²² The age at onset and gene-specific, age-related penetrance guide management to look for pheochromocytoma, paraganglioma, and nonchromaffin tumors in order to identify these neoplasms at an early stage, when they are resectable.

It is important that both patients and clinicians understand the concept of disease penetrance. Some studies provide penetrance figures that include all mutation carriers of a given gene, whereas others provide only data subsets for the clinical type of a syndrome such as MEN-2A and MEN-2B, for asymptomatic carriers, or for specific gene mutations. Common features of all susceptibility genes for pheochromocytoma or paraganglioma are age dependency and incomplete penetrance; thus, disease does not develop in all carriers of gene mutations. For example,

the mean penetrance of pheochromocytoma or paraganglioma in *RET* mutation carriers is 50% by the age of 44 years, whereas for carriers of mutations that increase susceptibility to von Hippel–Lindau disease type 2, the mean penetrance is 50% at the age of 52 years.^{55,57} *SDHA* mutation penetrance is 39% by the age of 40 years in probands, as compared with only 10% at the age of 70 years, with a lifetime disease penetrance of 1.7% in mutation carriers who are relatives of probands.^{73,85,86} Unlike *SDHA* mutation penetrance, *SDHB* mutation penetrance is similar in probands and relatives, at 22% by the age of 60 years, which also reflects lifetime penetrance, whereas lifetime penetrance is 8% for *SDHC* and 43% for *SDHD* by the age of 60 years.^{85,87} Penetrance estimations are pending for *MAX*, *TMEM127*, and *SDHAF2* and for the newer genes.

Postoperatively, asymptomatic patients should undergo annual biochemical testing and cross-sectional digital imaging of the operated area annually for at least 3 years. The intervals for subsequent surveillance imaging remain open to debate, but at most specialized centers, imaging is performed every 2 to 3 years (Table 1). Special guidelines are available for postoperative surveillance of patients with asymptomatic extraparaganglial tumors.^{59,60,88}

PREGNANCY IN PATIENTS
WITH PHEOCHROMOCYTOMA
OR PARAGANGLIOMA

Pheochromocytoma during pregnancy is regarded as one of the great challenges in medicine. Of paramount importance is interdisciplinary cooperation of obstetricians, endocrinologists, and surgeons. Pregnant patients have symptoms that are similar to those in patients who are not pregnant and may have had hypertension before becoming pregnant. The tumors can become symptomatic in any period of gestation. The diagnosis should be suspected clinically and confirmed biochemically. For imaging, ultrasonography and MRI without contrast material are recommended. Any nuclear medicine imaging is contraindicated. Pharmacologic treatment, mostly α -adrenergic blockade, is a challenge, since uteroplacental circulation must be adequately preserved. Endoscopic removal of the tumor is the option of choice and has been repeatedly

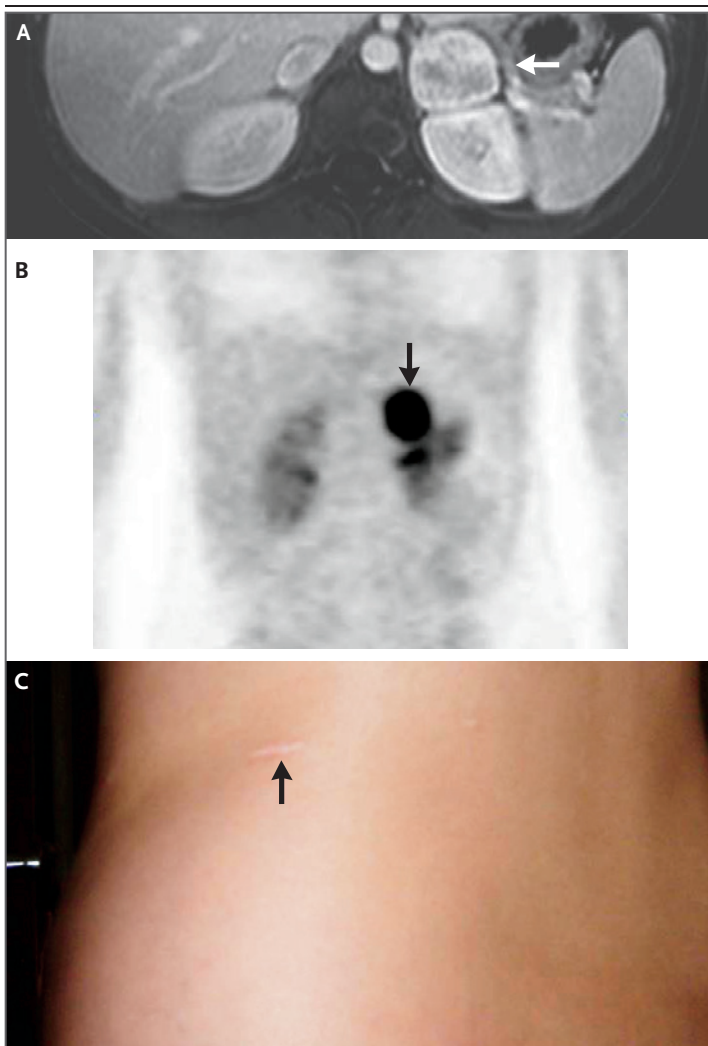


Figure 3. Presymptomatic Resection of a Juxta-Adrenal Paraganglioma after Mutation Identification in an 18-Year-Old Woman.

The patient's brother and father had pheochromocytoma. Genetic testing of the patient revealed germline mutation *VHL* c.298T→C. After the mutation was identified, the patient underwent *VHL*-informed MRI surveillance for von Hippel–Lindau disease, which revealed the juxta-adrenal paraganglioma. Five years later, the patient had an uneventful pregnancy and gave birth to a healthy child. In Panel A, MRI with gadolinium contrast medium (axial plane) shows the tumor (arrow). In Panel B, PET (frontal view) shows an ^{18}F -L-DOPA-avid paraganglioma (arrow). Panel C shows the 1-cm scar (arrow) after endoscopy with single-incision access was used to remove the tumor.

should be detected well before a pregnancy is planned. In this context, molecular genetics is an ideal tool. The proof of principle is shown in Figure 3, which presents the case of an 18-year-old woman who underwent resection of an asymptomatic paraganglioma after identification of a germline *VHL* mutation and had an uneventful pregnancy 5 years later.

CONCLUSIONS

More than 130 years have passed since the first patient with documented pheochromocytoma was described. The evolution in diagnosis and treatment has been dramatic. The clinical presentation has evolved from classic symptoms of paroxysmal hypertension, palpitations, and headaches to the incidental discovery of an adrenal mass on cross-sectional imaging and detection of asymptomatic pheochromocytomas on the basis of germline-mutation testing. The biochemical methods for case detection have become more sensitive and specific. Effective tumor localization has been facilitated by advances in cross-sectional imaging with CT and MRI and in functional imaging with ^{123}I -MIBG scintigraphy or ^{68}Ga -DOTATATE-PET-CT. The development of minimally invasive surgical techniques has improved operative outcomes. Individualized treatment and tumor surveillance have been remarkably advanced on the basis of the identification of 19 genes in which germline mutations cause pheochromocytoma or paraganglioma.

These endocrine neoplasms are rare, and many clinicians will never encounter them. Nevertheless, it is important for clinicians to remain vigilant and to suspect, confirm, localize, and resect these tumors, because the associated symptoms and hypertension are curable with surgical removal. If the tumor is not diagnosed and removed, there is a risk of lethal paroxysm and cardiac disorders. These tumors have malignant potential, and early resection is the way to prevent metastatic disease.

The evidence to date shows that more than 40% of patients presenting with pheochromocytoma or paraganglioma, irrespective of age at onset and family history, carry germline mutations. Ideally, all persons presenting with these neoplasms should undergo gene-panel testing in the context of genetic counseling, so that pro-

performed with good outcomes for mother and child; the preferred timing is the second trimester of pregnancy.⁸⁹⁻⁹¹ An alternative approach is medical management during pregnancy and resection of the pheochromocytoma several weeks after delivery. Ideally, however, such tumors

spective gene-specific management can be initiated proactively. If a patient is found to carry a mutation, all first-degree relatives should be offered mutation-specific testing. Although gene-informed surveillance, with early detection and cure, is a triumph of gene-informed precision care, it would be much more desirable to provide a targeted preventive therapy in order to avoid the risks of cumulative radiation doses and con-

trast medium associated with current tumor-surveillance strategies.

No potential conflict of interest relevant to this article was reported.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Tobias Krauss, M.D., Juri Ruf, M.D., Stephan Meckel, M.D., and Martin Walz, M.D., for providing clinical, radiographic, and histologic images; and Giuseppe Opocher, M.D., Francesca Schiavi, Ph.D., and Birke Bausch, M.D., for providing genetic data presented graphically.

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